

Backend Architecture Overview

This project uses a modular Express.js structure. Each user role (Student, Instructor, Admin and SuperAdmin) has its own set of controllers and routes under `src/modules/users`. Tutorials and their related resources live under `src/modules/users/tutorials`.

The valid roles available in the system are `Student`, `Instructor`, `Admin` and `SuperAdmin`. These values are referenced throughout the codebase and database migrations from a single shared constant.

Uploaded tutorial assets are stored in role specific folders:

```
backend/uploads/tutorials/admin/  
backend/uploads/tutorials/instructor/  
backend/uploads/tutorials/student/
```

The upload middleware automatically places files in these folders based on the authenticated user's role.